Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

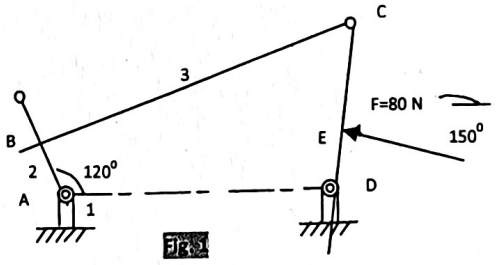
Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Program: B.Tech ME Vth Semester
Paper Name: Dynamics of Machines
Course Code: TME 503
Time: 1-1/2 Hour

MM: 50

Note: (i) All questions are compulsory.
(ii) Answer any **one** sub-questions among A and B in each main question.
(iii) Each question carries 10 marks.

Q.1	**10 Marks**	**CO1**
(a)	Define and explain the superposition theorem as applicable to a system of forces acting on a mechanism.	
	OR	
(b)	What do you mean by free body diagrams of a mechanism? Draw free body diagram of a slider crank mechanism.	
Q.2	**10 Marks**	**CO1**
(a)	(i)Define 'Inertia force' and 'Inertia torque' giving suitable examples. (ii)What do you mean by applied and constraint forces? Explain using a mechanism.	
	OR	
(b)	A four – link mechanism with the following dimensions is acted upon by a force of 80 N acting at an angle of 150^0 on the link DC (**Fig. 1**). AD= 500 mm, AB= 400mm,BC=1000 mm, DC= 750 mm, DE=350 mm. Determine the input torque T on the link AB for the static equilibrium of the mechanism for the given configuration.  Fig. 1	
Q.3	**10 Marks**	
(a)	What is a flywheel? What is its use? Find a relation for the coefficient of fluctuation of speed in terms of maximum fluctuation of energy and the kinetic energy of the	

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	flywheel at mean speed.	CO2
	OR	
(b)	Derive an expression for the angular acceleration of the connecting rod of a reciprocating engine.	
Q.4	**10 Marks**	
(a)	How are displacement, velocity and acceleration of the slider of a single slider crank mechanism determined analytically?	CO2
	OR	
(b)	Derive an expression for the correction torque to be applied to a crankshaft if the connecting rod of a reciprocating engine is replaced by two concentrated masses at the piston pin and the crank pin respectively.	
Q.5	**10 Marks**	
(a)	The piston diameter of an internal combustion engine is 125 mm and the stroke is 220 mm. The connecting rod is 4.5 times crank length and has a mass of 50 kg. The mass of the reciprocating parts is 30 kg. The centre of mass of the connecting rod is 170 mm from the crank pin centre and the radius of gyration about an axis through the centre of mass is 148 mm. The engine runs at 320 rpm. Find the magnitude and the direction of the inertia force and the corresponding torque on the crankshaft when the angle turned by the crank is 140^0 from the inner dead centre.	CO2
	OR	
(b)	In a single –acting four –stroke engine, the work done by the gases during the expansion stroke is three times the work done during the compression stroke. The work done during the suction and exhaust strokes is negligible. The engine develops 14 kW at 280 rpm. The fluctuation of speed is limited to 1.5 % of the mean speed on either side. The turning – moment diagram during the compression and the expansion strokes may be assumed to be triangular in shape. Determine the inertia of the flywheel.	

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